WORKED SOLUTION

STEP II 1993, Question 15

Probability and Statistics • Central Limit Theorem

STEP II 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q15

This question has a full guided treatment in *Beyond STEP*, Section 13.6, *Errors That Add Up*: https://beyondstep.keithmcnulty.org/ch13/errors-add-up.html

The key idea

The total error is a sum of many small, independent errors, one for each number. So work out the mean and variance of the error made on a *single* number, add them up over all the numbers, and use the Central Limit Theorem to treat the total as approximately normal. The whole contrast between the two computers comes from the means: one machine's errors cancel on average, while the other's all push the same way.

Solution

The error on one number

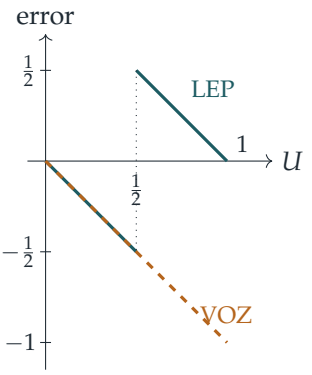
Let a number be $a = m + U$, where $m = \lfloor a \rfloor$ is an integer and U is its fractional part, with $U \sim \mathrm{U}[0,1)$. We define the error as (approximated value) $-$ (true value); the sign convention doesn't affect the magnitude.

LEP (rounding). If $U < \frac{1}{2}$ the number is replaced by m, an error of $-U$; if $U \geq \frac{1}{2}$ it is replaced by $m+1$, an error of $1-U$. So the error is

$$X = \begin{cases} -U, & 0 \leq U < \frac{1}{2}, \\ 1-U, & \frac{1}{2} \leq U < 1, \end{cases}$$

which takes each value in $(-\frac{1}{2}, \frac{1}{2}]$ with constant density: X is uniform on $[-\frac{1}{2}, \frac{1}{2}]$ (the sawtooth in the figure).

VOZ (truncation). The number is replaced by m, so the error is $Y = -U$, uniform on $(-1, 0]$.



A uniform distribution on an interval of length 1 has variance $\frac{1}{12}$, wherever the interval sits. So

$$\mathrm{E}(X) = 0, \quad \mathrm{Var}(X) = \tfrac{1}{12}; \qquad \mathrm{E}(Y) = -\tfrac{1}{2}, \quad \mathrm{Var}(Y) = \tfrac{1}{12}.$$

(For the record, $\mathrm{Var}(U) = \int_0^1 u^2\,\mathrm{d}u - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.)

Adding 1500 numbers

The error in the answer is the sum of the errors on the individual numbers, since the computer adds the approximated values exactly. The fractional parts are independent, so the individual errors are independent and their variances add. For $n = 1500$ numbers,

$$\mathrm{Var}(\text{total error}) = \frac{1500}{12} = 125, \qquad \text{standard deviation } \sqrt{125} = 5\sqrt{5} \approx 11.18.$$

By the Central Limit Theorem, a sum of 1500 independent, identically distributed errors is very nearly normal.

LEP. The total error E_L is approximately $\mathrm{N}(0, 125)$, so

$$\begin{aligned}\mathrm{P}(|E_L| > 15) &\approx \mathrm{P}\Big(|Z| > \frac{15}{\sqrt{125}}\Big) = \mathrm{P}(|Z| > 1.342)\\ &= 2\big(1 - \Phi(1.342)\big) \approx 2(1 - 0.9102) \approx 0.18.\end{aligned}$$

No continuity correction is needed: the errors are continuous random variables, not integers.

VOZ. The total error E_V is approximately $\mathrm{N}(-750, 125)$. The error is only at most 15 in size if $E_V \geq -15$, which is

$$\frac{-15 - (-750)}{\sqrt{125}} = \frac{735}{11.18} \approx 66$$

standard deviations above the mean. That probability is utterly negligible, so

$$\mathrm{P}(|E_V| > 15) \approx 1.$$

Truncation always errs downwards, by $\frac{1}{2}$ on average, so after 1500 numbers VOZ is out by about 750.

Ask yourself Both machines make errors of the same variance. Why are the answers so different?

The spread of the total error is the same, ± 11 or so, but its *centre* is not. LEP's errors are centred at 0, so the total wanders around 0. VOZ's errors all have mean $-\frac{1}{2}$, so the bias grows in proportion to n, while the random spread grows only like $\sqrt{n}$. The bias wins easily.

How many additions for LEP?

Suppose LEP adds n numbers. Its total error is approximately $\mathrm{N}(0, \frac{n}{12})$, and we need

$$\mathrm{P}(|E_L| < 10) = 2\Phi\Big(\frac{10}{\sqrt{n/12}}\Big) - 1 \;\geq\; 0.9 \quad\Longleftrightarrow\quad \Phi\Big(\frac{10}{\sqrt{n/12}}\Big) \geq 0.95.$$

Since Φ is increasing and $\Phi(1.6449) = 0.95$, this is

$$\frac{10}{\sqrt{n/12}} \geq 1.6449 \iff n \leq 12\Big(\frac{10}{1.6449}\Big)^2 \approx 443.5.$$

The probability decreases as n increases, so it stays at least 0.9 for $n \leq 443$ and first drops below 0.9 at $n = 444$. So LEP can add **443** numbers. (If you count an "addition" as one use of the $+$ sign, adding 443 numbers takes 442 of them; either reading is fine if you say which you mean.)

Checks

- *Is* 1.342 *believable?* A total error of 15 is a little over one standard deviation (11.18), and about 18% of a normal distribution lies more than 1.34 standard deviations from the mean. That's about right.
- *Numerically.* A simulation of 10^5 batches of 1500 rounded numbers gave $\mathrm{P}(|E_L| > 15) \approx 0.179$. The exact distribution of the sum of uniforms gives 0.1797, against 0.1797 from the normal approximation. For the second part the exact probability is 0.90019 at $n = 443$ and 0.89981 at $n = 444$, confirming the cut-off.

Common slips

- Taking LEP's error to be uniform on $[0, \frac{1}{2}]$: rounding can go up or down, so the error is uniform on $[-\frac{1}{2}, \frac{1}{2}]$.
- Forgetting VOZ's mean of $-\frac{1}{2}$ per number and so giving it the same answer as LEP.
- Using $\Phi(z) = 0.9$ instead of 0.95 in the last part: the event $|E| < 10$ is two-sided, so each tail gets 0.05.

Beyond STEP

This is the classic analysis of rounding error in floating-point arithmetic: unbiased rounding makes the accumulated error grow like $\sqrt{n}$, while a biased rule makes it grow like n. That is why the IEEE 754 standard uses "round half to even" rather than always rounding halves upwards. The sum of n uniforms has an exact distribution (the Irwin–Hall distribution), and seeing how quickly it becomes normal is a good way into the Central Limit Theorem; Grimmett and Welsh, *Probability: An Introduction*, proves the theorem and discusses it.

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